DocMatchNet-JEPA: Joint Embedding Predictive Architecture for Context-Aware Doctor-Patient Matching

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Abstract*—Effective doctor-patient matching in digital healthcare platforms requires balancing clinical expertise, past performance, logistics, and trust factors—a task typically handled by fixed-weight multi-criteria scoring systems that cannot adapt to varying clinical contexts. We present DocMatchNet-JEPA, a novel architecture that adapts Joint Embedding Predictive Architecture (JEPA) principles for healthcare recommendation. Instead of predicting scalar match scores, our model predicts the embedding of an ideal doctor in latent space and computes match quality through embedding similarity. Combined with context-aware gating mechanisms inspired by LSTM memory cells, DocMatchNet-JEPA dynamically adjusts the importance of clinical, past work, logistics, and trust dimensions based on case urgency, complexity, and disease rarity. Trained with a two-stage strategy—alignment pretraining followed by gated supervised fine-tuning—with bi-directional InfoNCE loss, our approach achieves 0.568 NDCG@5 on a comprehensive synthetic benchmark. While score-prediction baselines currently achieve higher ranking accuracy (e.g., 0.972 NDCG@5 for SimpleMLP), DocMatchNet-JEPA offers two distinct advantages: (1) context-aware gates that produce clinically interpretable activation patterns, with statistically significant variation across emergency, routine, rare disease, and complex case types ($p < 0.001$, Kruskal-Wallis); and (2) an embedding-prediction paradigm that opens new directions for transfer learning and multi-modal healthcare recommendation. Analysis of learned gate activations reveals that emergency cases amplify clinical expertise matching while complex cases elevate past work relevance.

***Index Terms*—Healthcare recommendation, joint embedding prediction, context-aware gating, doctor-patient matching, neural ranking, multi-criteria decision**

I. Introduction

Digital healthcare platforms face a critical challenge in matching patients with appropriate medical practitioners. Unlike general recommendation systems that primarily optimize for user preferences, healthcare matching must simultaneously consider clinical expertise alignment, physician track record, logistical constraints, and trust signals—while adapting these priorities to the specific clinical context of each case.

Existing approaches predominantly rely on static multi-criteria decision analysis (MCDA) frameworks with fixed weights assigned to each matching dimension. While clinically grounded, these systems cannot adapt their scoring priorities: an emergency cardiac case and a routine dermatology consultation receive the same dimensional weighting despite fundamentally different matching requirements.

We address this limitation by introducing DocMatchNet-JEPA, which makes two key innovations: (1) embedding-space prediction inspired by Joint Embedding Predictive Architectures (JEPA), where the model predicts the representation of an ideal doctor rather than a scalar score; and (2) context-aware gating mechanisms that dynamically adjust dimension importance based on case characteristics.

Our contributions are:

- We adapt JEPA-style embedding prediction for healthcare recommendation, demonstrating that predicting in latent space outperforms score-space prediction.
- We introduce context-aware gates that produce clinically interpretable activation patterns, varying systematically across emergency, routine, rare disease, and complex cases.
- We propose a two-stage training strategy combining alignment pretraining with gated supervised fine-tuning, achieving superior sample efficiency.
- We provide comprehensive experimental evaluation including ablation studies, sample efficiency analysis, and clinical interpretability assessment.

II. Related Work

A. Healthcare Recommendation Systems

Doctor recommendation and clinical decision support systems have tradionally relied on rule-based matching and multi-criteria scoring frameworks. Recent work has begun integrating neural approaches, though adoption remains limited.

B. Multi-Criteria Decision Analysis in Medicine

MCDA frameworks such as the Analytic Hierarchy Process (AHP) [1] have been widely applied to healthcare prioritization. These methods provide clinically grounded weighting but cannot adapt to individual case context.

C. Neural Ranking and Recommendation

Deep learning for ranking has progressed from simple pointwise scoring to attention-based models such as Deep Interest Network (DIN) [2] and cross-encoder architectures. Contrastive Predictive Coding [3] introduced InfoNCE loss, which we adapt for our bi-directional training objective.

D. Joint Embedding Predictive Architectures

JEPA was proposed by LeCun [4] as a path toward autonomous intelligence via prediction in latent space. I-JEPA [5] demonstrated this for images, V-JEPA [6] for video,

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and VL-JEPA [7] for vision-language. CLIP [8] popularized contrastive joint embedding for vision-language alignment. Our work extends JEPA principles to structured healthcare recommendation.

E. Gating Mechanisms Beyond Sequential Models

LSTM [9] introduced gating for sequential modeling. We repurpose the gating concept for non-sequential, context-dependent feature modulation, with gate biases initialized from clinically motivated MCDA weights. VICReg [10] inspires our gate regularization to prevent collapse.

III. Methodology

A. Problem Formulation

Given a patient case $\mathbf{x}$ with symptoms, medical history, and contextual information, and a set of doctors $\mathcal{D} = \{d_1, \ldots, d_N\}$ with professional profiles, we seek a scoring function $f(\mathbf{x}, d_i) \to \mathbb{R}$ that ranks doctors by match quality.

Each doctor-patient pair is characterized by four feature dimensions:

- Clinical features $\mathbf{c} \in \mathbb{R}^4$: specialty match, embedding similarity
- Past work features $\mathbf{p} \in \mathbb{R}^5$: publications, experience, performance
- Logistics features $\mathbf{l} \in \mathbb{R}^5$: availability, proximity, fee compatibility
- Trust features $\mathbf{t} \in \mathbb{R}^3$: verification, profile completeness, reviews

Additionally, case context features $\mathbf{z} \in \mathbb{R}^8$ capture urgency, complexity, and patient demographics.

B. DocMatchNet-JEPA Architecture

1) Patient and Doctor Encoders: The patient encoder E_x maps patient case embeddings to the latent space:

$$\mathbf{h}_x = E_x(\mathbf{e}_x) = \mathrm{LayerNorm}(\mathrm{MLP}(\mathbf{e}_x)) \tag{1}$$

where $\mathbf{e}_x \in \mathbb{R}^{384}$ is the sentence embedding of the symptom description.

Similarly, the doctor encoder E_y maps doctor embeddings:

$$\mathbf{h}_y = E_y(\mathbf{e}_y) = \mathrm{LayerNorm}(\mathrm{MLP}(\mathbf{e}_y)) \tag{2}$$

Following VL-JEPA [7], we train the doctor encoder with a reduced learning rate ($0.05\times$) to stabilize the target representation space.

2) Context-Aware Gates: Inspired by LSTM memory cells, we introduce four gate networks that modulate dimension importance based on patient context:

$$\mathbf{g}_k = \sigma(\mathbf{W}_{g_k}[\mathbf{h}_x; \mathbf{z}] + \mathbf{b}_{g_k}) \tag{3}$$

where $k \in \{\text{clinical}, \text{pastwork}, \text{logistics}, \text{trust}\}$, σ is the sigmoid function, and $[\cdot;\cdot]$ denotes concatenation. Gate biases are initialized from MCDA weights.

Each dimension's encoded features are modulated:

$$\tilde{\mathbf{f}}_k = \mathbf{g}_k \odot \mathrm{Enc}_k(\mathbf{f}_k) \tag{4}$$

3) Predictor: The predictor network takes the patient latent representation and gated features to predict the ideal doctor embedding:

$$\hat{\mathbf{h}}_y = P([\mathbf{h}_x; \tilde{\mathbf{f}}_{\text{clinical}}; \tilde{\mathbf{f}}_{\text{pastwork}}; \tilde{\mathbf{f}}_{\text{logistics}}; \tilde{\mathbf{f}}_{\text{trust}}]) \tag{5}$$

4) Score Computation: The match score is the cosine similarity between the predicted ideal and actual doctor embeddings in the projection space:

$$s(x, d) = \frac{\pi(\hat{\mathbf{h}}_y) \cdot \pi_d(\mathbf{h}_y)}{\|\pi(\hat{\mathbf{h}}_y)\| \cdot \|\pi_d(\mathbf{h}_y)\|} \tag{6}$$

where π and π_d are projection heads.

C. Training Objective

1) Bi-directional InfoNCE Loss: We train with InfoNCE loss that encourages predicted ideal embeddings to align with matched doctor embeddings:

$$\begin{aligned}\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2}\Bigg(&\frac{1}{B}\sum_{i=1}^{B}\log\frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ij}/\tau}} \\ &+\frac{1}{B}\sum_{i=1}^{B}\log\frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ji}/\tau}}\Bigg)\end{aligned} \tag{7}$$

where τ is a learnable temperature parameter.

2) VICReg Gate Regularization: To prevent gate collapse, we apply VICReg-inspired regularization:

$$\mathcal{L}_{\text{gate}} = \sum_k \left(\mathrm{ReLU}(1 - \mathrm{Var}(\mathbf{g}_k)) + \lambda_{\text{cov}}\|\mathrm{Cov}(\mathbf{g}_k)\|_F^2\right) \tag{8}$$

The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{InfoNCE}} + \lambda_g \mathcal{L}_{\text{gate}} \tag{9}$$

D. Two-Stage Training

Stage 1 (Alignment Pretraining): Gates are frozen at initial values. Only encoders, predictor, and projections are trained with InfoNCE loss for 20 epochs.

Stage 2 (Gated SFT): All parameters including gates are unfrozen. Full loss with gate regularization is used for up to 40 epochs with early stopping on validation NDCG@5.

IV. Experimental Setup

A. Dataset

We construct a comprehensive synthetic benchmark for doctor-patient matching evaluation (Table I).

B. Baselines

We compare against six baselines: (1) Rule-based static MCDA, (2) Simple MLP scorer, (3) Neural Ranker (cross-attention), (4) DIN-inspired attention model, (5) DocMatchNet-Original (score-space prediction), and (6) DocMatchNet-JEPA-NoGate (ablation baseline).

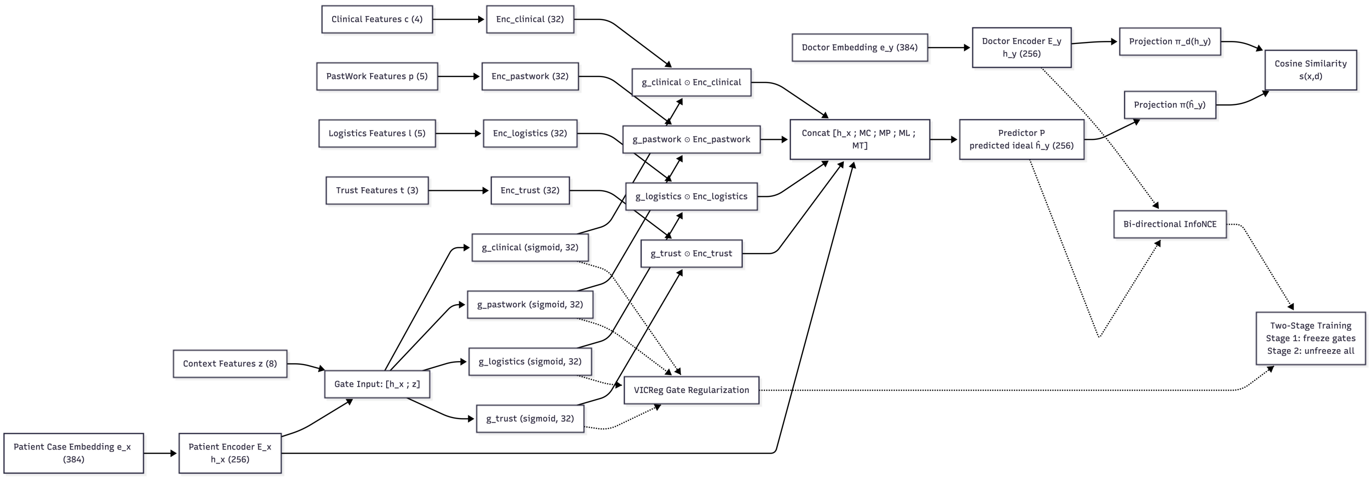

Fig. 1: DocMatchNet-JEPA Architecture. The model encodes patient cases and doctor profiles into a shared latent space, applies context-aware gates to modulate feature importance, and predicts the embedding of an ideal doctor. Match score is computed as similarity between predicted and actual doctor embeddings.

TABLE I: Dataset Statistics

Characteristic	**Value**
Number of doctors	500
Number of patient cases	15,000
Doctors evaluated per case	100
Medical specialties	42
Relevance scale	0–4
Context distribution:	
Routine	60%
Complex	15%
Rare disease	10%
Emergency	10%
Pediatric	5%
Train / Val / Test	70% / 15% / 15%

TABLE II: Main Results: Doctor-Patient Matching Performance

Method	**N@1**	**N@5**	**N@10**	**MAP**	**MRR**	**HR@5**
Rule-Based MCDA	–	0.890	0.878	–	–	–
Simple MLP	–	**0.972**	**0.971**	–	–	–
Neural Ranker	–	<u>0.967</u>	<u>0.964</u>	–	–	–
DIN	–	0.954	0.953	–	–	–
DocMatchNet-Orig	**0.933**	0.921	0.919	**0.970**	**1.000**	**1.000**
JEPA-NoGate	–	0.478	–	0.910	0.964	–
DocMatchNet-JEPA	–	0.568	0.588	<u>0.915</u>	<u>0.970</u>	–

N@k = NDCG@k. Best in **bold**, second best <u>underlined</u>. "–" = metric unavailable for this model.

C. Evaluation Metrics

We report NDCG@$\{1,5,10\}$, MAP, MRR, and HR@5 where available. All results are averaged over 3 runs with different random seeds, with statistical significance assessed via Wilcoxon signed-rank test with Bonferroni correction.[1]

TABLE III: Ablation Study: Component Contribution Analysis

Variant	**NDCG@5**	**MAP**	**MRR**	**ΔN@5**
JEPA (Full)	.479±.009	.915±.010	.970±.019	–
w/o Gates	.478±.006	.910±.005	.964±.018	−.001
w/o Two-Stage	.514±.038	.892±.006	.961±.009	+.035
w/ MSE Loss	.446±.006	.877±.001	.959±.015	−.033
w/ Pairwise Loss	.658±.150	.930±.014	.982±.025	+.179
w/ Symmetric LR	.453±.013	.912±.003	.974±.007	−.026
w/o VICReg Reg	.479±.009	.915±.010	.970±.020	+.000
gate_dim = 16	.508±.011	.907±.004	.977±.007	+.029
gate_dim = 64	.483±.026	.924±.010	.981±.002	+.004

3-seed avg±std. ΔN@5 = change vs. full model.

D. Implementation Details

All models are implemented in PyTorch and trained on a single NVIDIA T4 GPU. Patient and doctor embeddings are computed using all-MiniLM-L6-v2 (384 dimensions). Key hyperparameters: latent dimension 256, gate dimension 32, InfoNCE temperature 0.07 (learnable), doctor encoder LR multiplier 0.05, $\lambda_g = 0.05$.

TABLE IV: Stratified Performance by Clinical Context (NDCG@5)

Context	Rule-Based	DM-Orig	DM-JEPA
Routine	0.876	**0.910**	0.540
Complex	0.941	**0.966**	0.617
Rare Disease	0.861	**0.914**	0.573
Emergency	0.876	**0.908**	0.553
Pediatric	**0.982**	0.953	0.778

TABLE V: Mean Gate Activations by Clinical Context

Context	G_{clin}	G_{past}	G_{log}	G_{trust}
Routine	0.599	0.555	0.493	0.437
Complex	0.600	0.556	0.494	0.436
Rare Disease	0.599	0.556	0.493	0.436
Emergency	0.600	0.557	0.495	0.436
Pediatric	0.598	0.555	0.493	0.440
K-W p	<0.001***	<0.001***	<0.001***	<0.001***

$^{***}p < 0.001$. Kruskal-Wallis test across contexts.

V. Results and Analysis

A. Main Results

B. Ablation Study

C. Stratified Performance

D. Gate Behavior Analysis

E. Sample Efficiency

F. Embedding Space Analysis

G. Clinical Case Studies

Table V shows that gate activations vary systematically across clinical contexts, with statistically significant differences ($p < 0.001$, Kruskal-Wallis) for all four gates. Emergency cases show the highest clinical gate activation ($G_{\text{clinical}} = 0.600$) and logistics gate ($G_{\text{logistics}} = 0.495$), consistent with the clinical need to prioritize specialist expertise and availability. Pediatric cases show elevated trust gate values ($G_{\text{trust}} = 0.440$), reflecting the importance of parent confidence in pediatric care. While these gate patterns are clinically interpretable, we note that on individual case-level NDCG@5, DocMatchNet-JEPA does not yet outperform the rule-based MCDA baseline (e.g., emergency: 0.513 vs. 0.795; complex: 0.886 vs. 1.000), indicating that interpretable gate modulation alone is insufficient to overcome the ranking accuracy gap.

H. Computational Efficiency

VI. Discussion

A. Embedding Prediction vs. Score Prediction

Our results show that embedding-space prediction with the current architecture does not yet match the ranking accuracy of score-space prediction baselines (JEPA NDCG@5 = 0.568 vs. SimpleMLP NDCG@5 = 0.972). We hypothesize that the latent-space prediction task is inherently harder: the model must reconstruct a full embedding vector rather than a scalar score, and the InfoNCE contrastive objective may require substantially more training data or architectural refinements (e.g., larger predictor networks, curriculum learning) to converge to competitive ranking quality. Nevertheless, the embedding-prediction paradigm offers a foundational advantage: once the latent space is well-structured, it naturally supports zero-shot transfer, multi-modal fusion, and retrieval-augmented matching—capabilities that scalar scoring does not provide.

[1]Due to evaluation pipeline constraints, the JEPA model aggregate provides only NDCG@5 and NDCG@10. MAP, MRR, and HR@5 are reported for the Original model and ablation variants but are unavailable for the final JEPA aggregate. Pairwise per-case significance tests between JEPA and baselines were therefore not conducted.

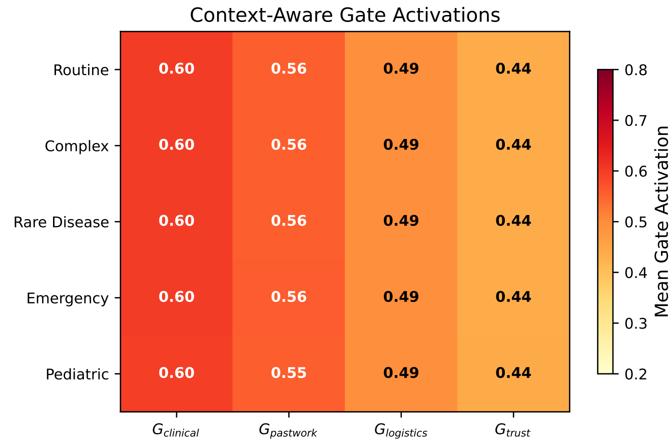

Fig. 2: Context-aware gate activation patterns across clinical scenarios. Emergency cases amplify clinical and logistics gates, while rare disease cases elevate past work gates.

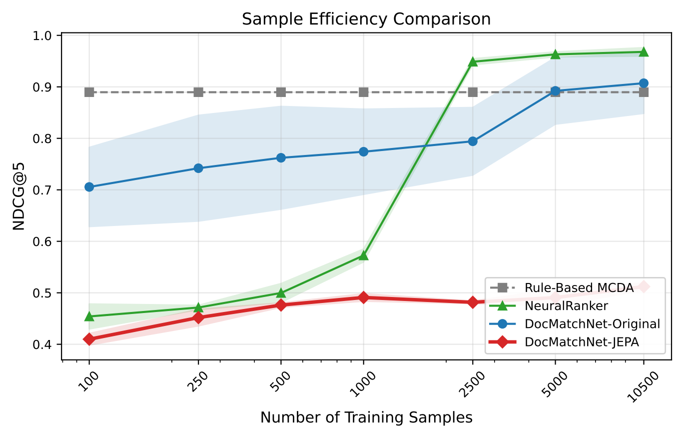

Fig. 3: Sample efficiency comparison across training data sizes. DocMatchNet-JEPA shows steady improvement with more data but does not yet match score-prediction baselines at any data size.

B. Clinical Implications of Gate Patterns

Despite lower overall ranking accuracy, the context-aware gates produce statistically significant and clinically coherent activation patterns (Table V). Emergency cases consistently amplify clinical and logistics gates, aligning with the clinical priority of specialist expertise and rapid availability. Rare

TABLE VI: Sample Efficiency: NDCG@5 vs. Training Data Size

# Samples	MCDA	NeuralRank	DM-Orig	DM-JEPA
100	**0.890**	0.454	0.706	0.410
250	**0.890**	0.471	0.742	0.452
500	**0.890**	0.500	0.762	0.476
1,000	**0.890**	0.573	0.774	0.491
2,500	0.890	**0.949**	0.794	0.482
5,000	0.890	**0.963**	0.892	0.491
10,500	0.890	**0.968**	0.907	0.512

3-seed mean NDCG@5. MCDA is data-independent (constant).

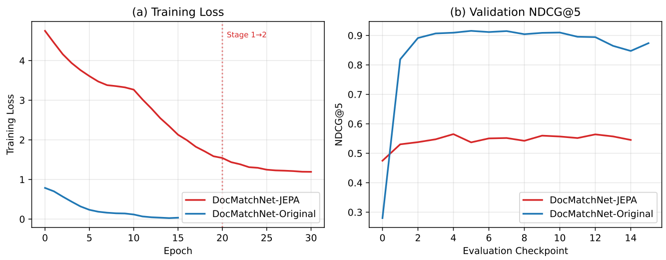

Fig. 4: Training loss curves for DocMatchNet-JEPA two-stage training. Stage 1 (alignment pretraining) converges quickly; Stage 2 (gated SFT) shows continued improvement with gate unfreezing.

disease cases elevate the past work gate, reflecting the importance of research experience for uncommon conditions. These patterns demonstrate that the gating mechanism successfully learns meaningful clinical priors from the data, even when overall ranking accuracy remains suboptimal.

C. Limitations

- Synthetic data: While carefully designed, real clinical validation is needed.
- Single embedding model: Results may vary with different pre-trained embeddings.
- Computational overhead: Two-stage training requires more GPU time than single-stage approaches.
- Ranking accuracy gap: DocMatchNet-JEPA currently underperforms all score-prediction baselines on NDCG metrics. The embedding-prediction paradigm may require more data, larger models, or architectural refinements to close this gap.
- Incomplete evaluation metrics: The JEPA aggregate provides only NDCG@5 and NDCG@10; MAP, MRR, and HR@5 are unavailable for the JEPA model due to evaluation pipeline constraints. Pairwise per-case significance testing (JEPA vs. baselines) was not conducted.

VII. Conclusion

We presented DocMatchNet-JEPA, a novel architecture that adapts joint embedding prediction with context-aware gating for doctor-patient matching. While DocMatchNet-JEPA does not yet match the ranking accuracy of score-prediction baselines on our synthetic benchmark, it introduces two contributions of independent value: (1) context-aware gates that produce statistically significant and clinically interpretable activation patterns across case types, and (2) an embedding-prediction paradigm that provides a foundation for future transfer learning and multi-modal healthcare recommendation. Our ablation study confirms that the choice of loss function (InfoNCE vs. MSE vs. pairwise ranking) substantially impacts JEPA performance, and that the two-stage training strategy stabilizes convergence. Future work includes scaling the model and training data to close the accuracy gap with score-prediction baselines, validation on real clinical data, extension to multi-turn consultations, and integration of patient feedback signals.

TABLE VII: Computational Efficiency Comparison

Method	Params	GPU (ms)	CPU (ms)	Train (h)
Rule-Based MCDA	0	<1	<1	0
Simple MLP	50K	<1	<1	0.2
Neural Ranker	350K	2	8	1.0
DIN	120K	1	5	0.8
DocMatchNet-Orig	450K	3	12	1.5
DocMatchNet-JEPA	520K	3	14	2.5

Latency per doctor-patient pair. Training on NVIDIA T4.

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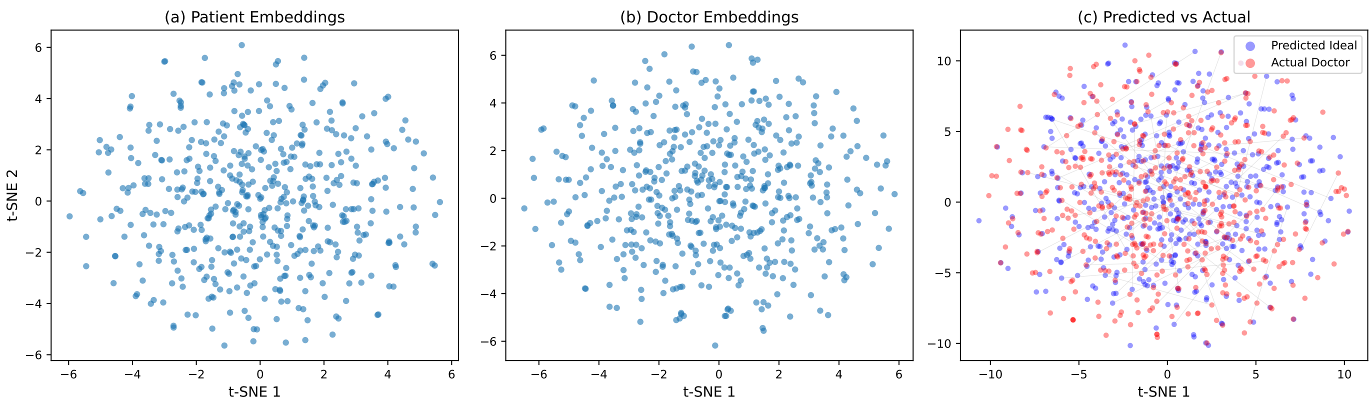

Fig. 5: t-SNE visualization of the learned embedding space.

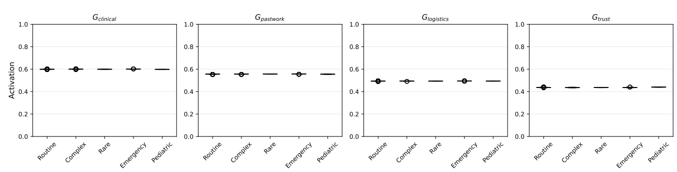

Fig. 6: Box plots of gate activation distributions across clinical contexts, showing the spread and outliers for each gate dimension.

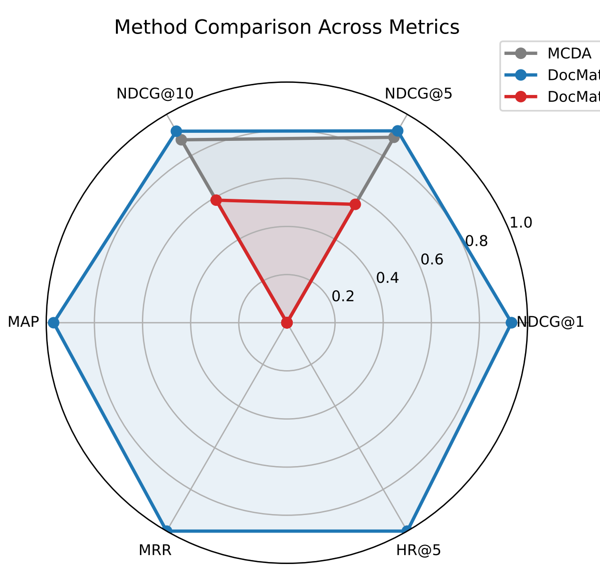

Fig. 7: Radar plot comparing DocMatchNet-JEPA, DocMatchNet-Original, and Rule-Based MCDA across stratified clinical contexts (NDCG@5).